

BLE Angle-of-Arrival Navigation System

Extending WiFi robot navigation with BLE localization
using the u-blox XPLR-AOA-3

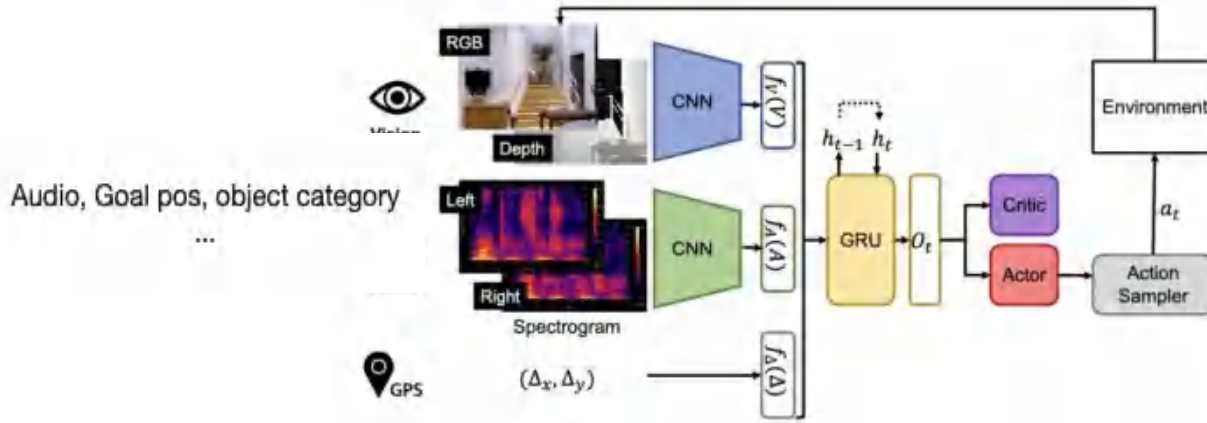
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Group 5

May 5, 2026

Motivation: Robot navigation using wireless signal

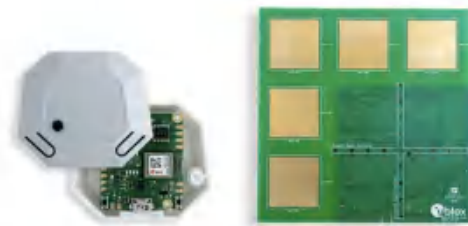
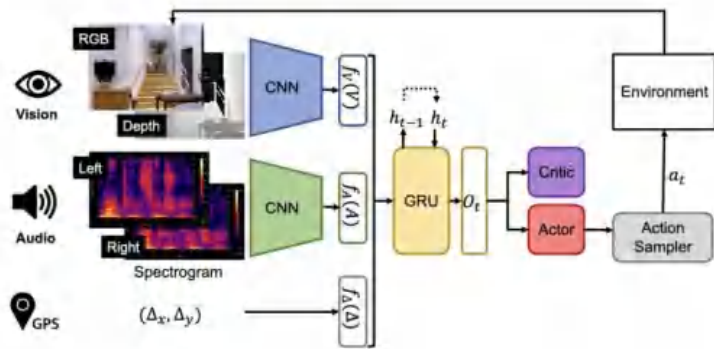
- Navigation: In an unmapped environment, an agent must navigate to a goal object within limited actions.



- Vision-only solutions: (1) category-level goal brings ambiguity; (2) NLoSgoal: requires exploration before navigation.
- can we rethink navigation from a signal perspective?

Motivation: Extending WiFi Navigation with BLE Signals

- **Advantages of wireless signals:** Enable instance-level identification via MAC addresses and allow localization even when the target is not visually observable.
- Current research: The existing system explores **WiFi-based signal sensing** for robot navigation.
- **Our contribution:** Integrate **BLE localization** using the **u-blox XPLR-AOA-3 AoA platform** to extend WiFi sensing into a **multimodal signal-based navigation framework**.



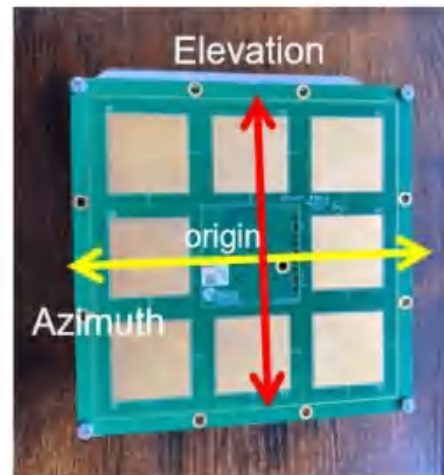
XPLR-AOA-3 Setup

First-time Setup

- ① **Install s-center classic** (u-blox website, NINA-B series)
- ② **Flash C209 firmware:** remove battery, connect Micro-USB, hold SW2+RESET to enter DFU mode, use s-center Software Update with `release_for_dfu_boot .zip` from GitHub
- ③ **Configure EVB-ANT-1:** J2 jumper on F (FTDI) pins, J12 no jumper, connect USB-C to PC
- ④ **Open s-center on COM11** (115200, CTS/RTS) > IoT > Angle of Arrival > enable all checkboxes > Set
- ⑤ **Install CR2032 battery in C209**

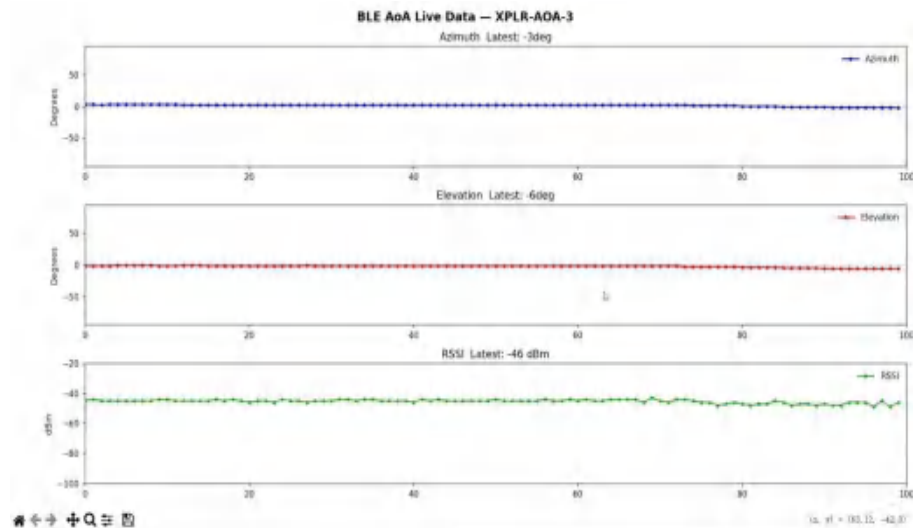
Subsequent Use

- ① **EVB-ANT-1:** plug in USB-C, open s-center on COM11 (115200, CTS/RTS), IoT > Angle of Arrival > Resume
- ② **C209 tag:** insert CR2032 battery or use USB. Tag broadcasts automatically. Tag ID appears in s-center dropdown within seconds
- ③ **Data logging:** close COM11 in s-center, run `ble_aoa_logger.py` to record Azimuth, Elevation, RSSI to CSV with live plots



In s-center, we can obtain the azimuth angle and the elevation angle.

Azimuth and elevation angle detect



From the origin, move in the four directions: forward, backward, left, and right
Visualize the angle variations and save the data

Setup for elevation and azimuth



figure 1

figure 2



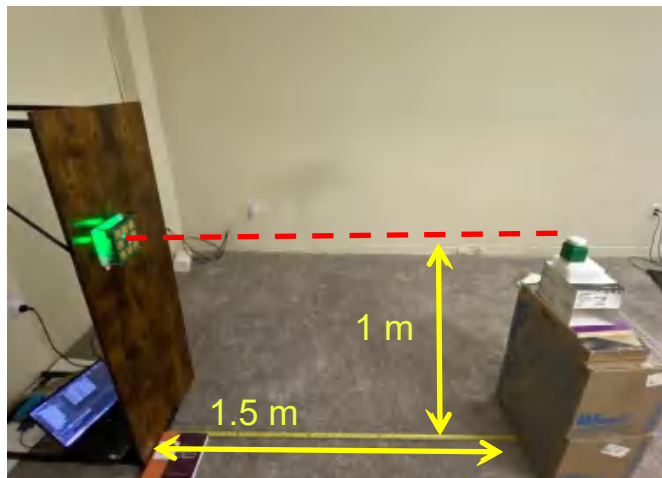
elevation



azimuth

- The definitions of elevation and azimuth are **fixed** relative to the antenna board
- No matter how I rotate the antenna board, these **two axes rotate with the board**
- Since the **tag's position** is easier to control and measure in the **horizontal** direction, the **elevation axis** is made horizontal in Figure 1, and the **azimuth axis** is made horizontal in Figure 2.

Setup for LOS and NLOS



Placing the tag and the antenna board **too close to the ground** will reduce accuracy

placed both of them 1 m above the ground and moved the tag **horizontally**

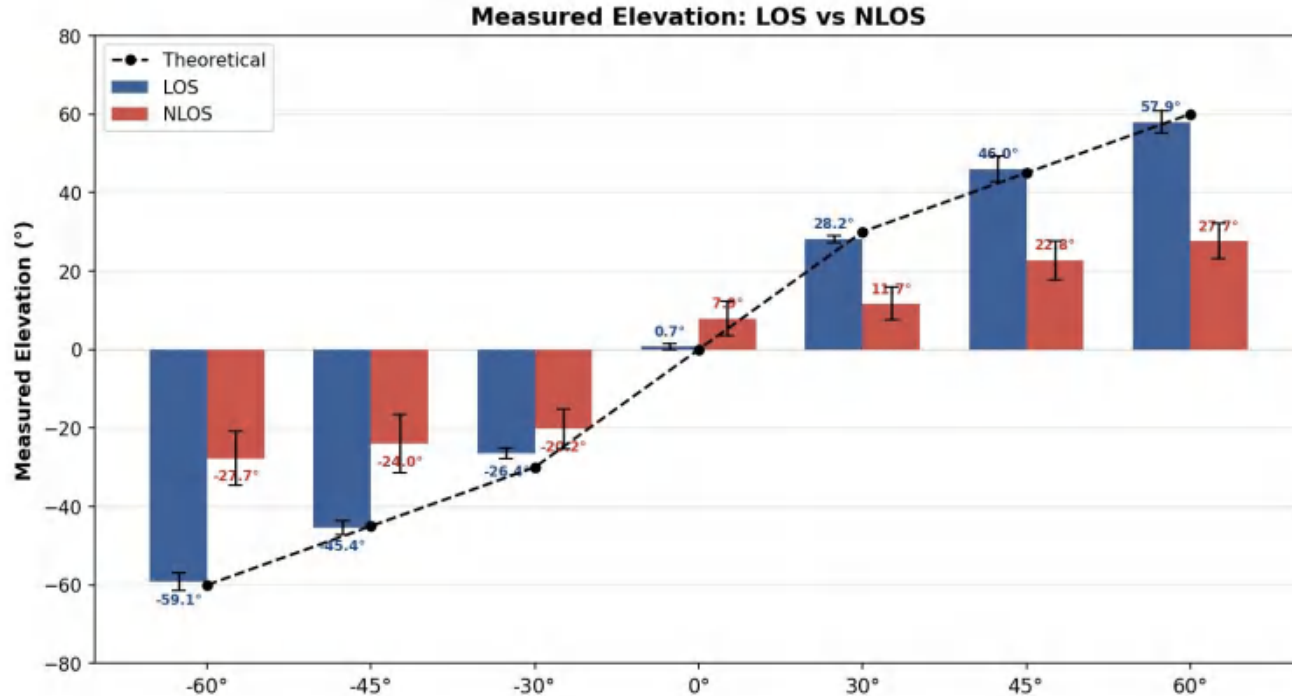


Test $0^\circ \pm 30^\circ \pm 45^\circ \pm 60^\circ$



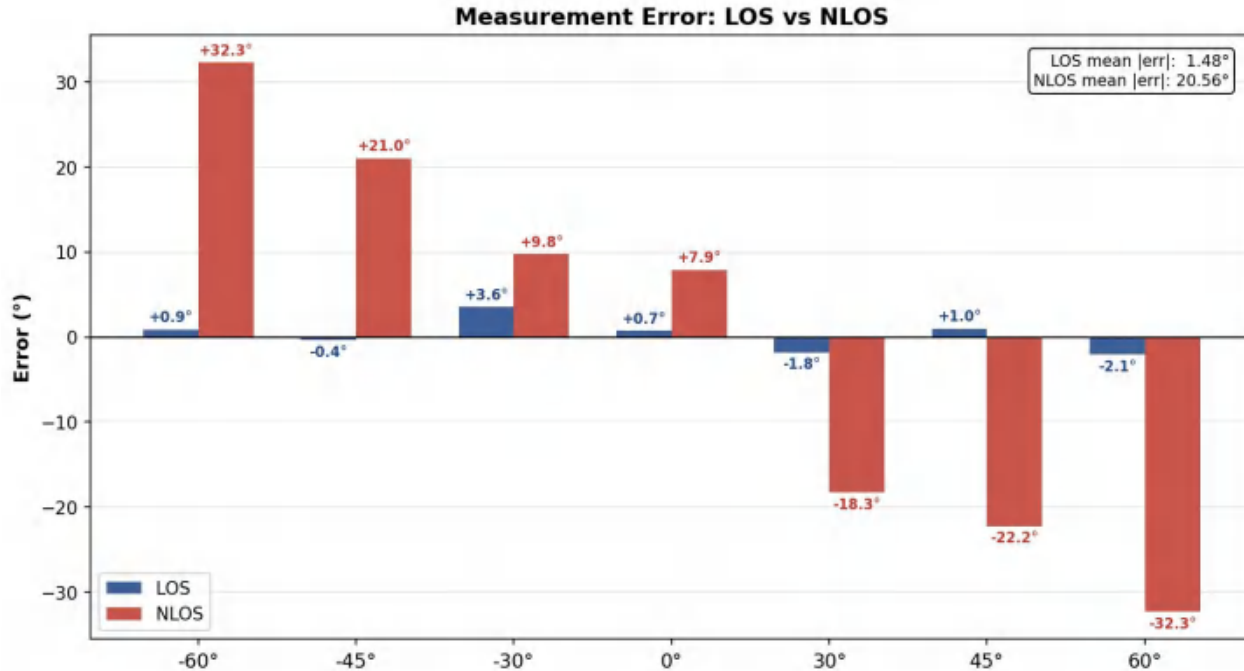
NLOS is achieved by human body obstruction

BLE elevation LOS/NLOS test



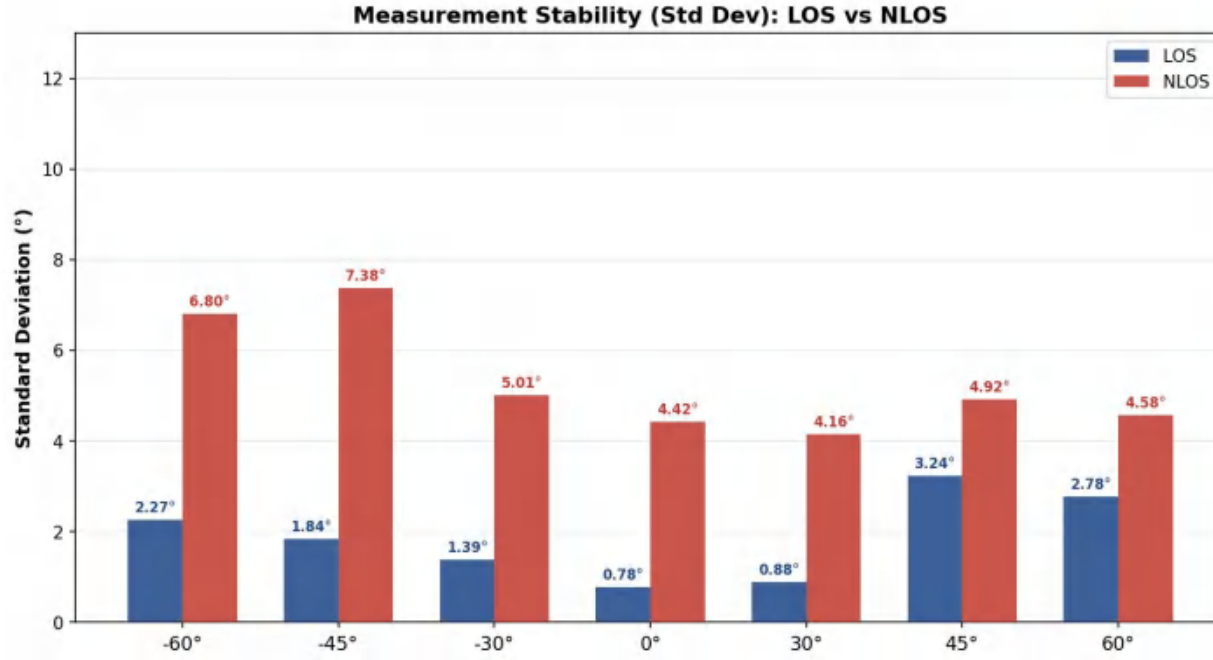
- Under NLOS conditions, the measured values **shift toward 0°** for both positive and negative angles

BLE elevation LOS/NLOS test



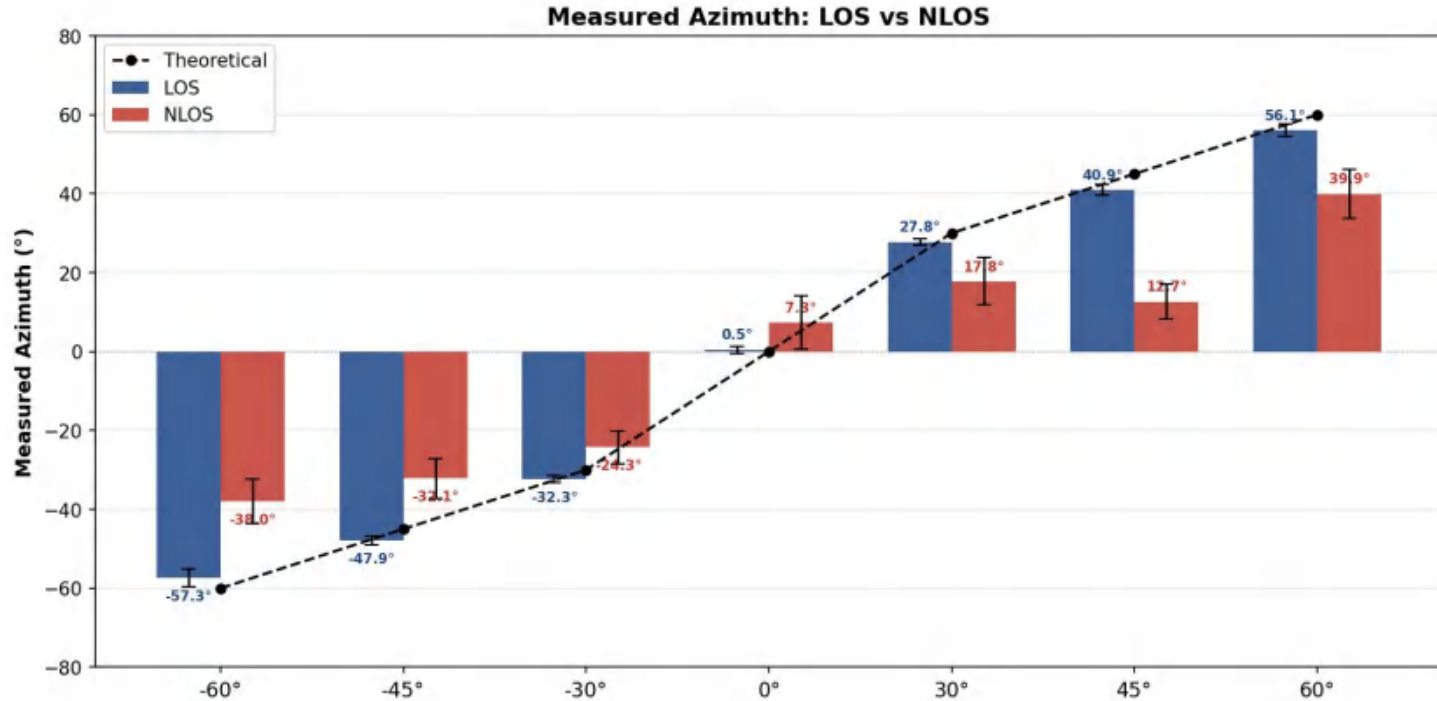
- The LOS error for all angles is within $\pm 4^\circ$, indicating stable measurements.
- The NLOS error increases with the angle, significantly reducing measurement accuracy

BLE elevation LOS/NLOS test



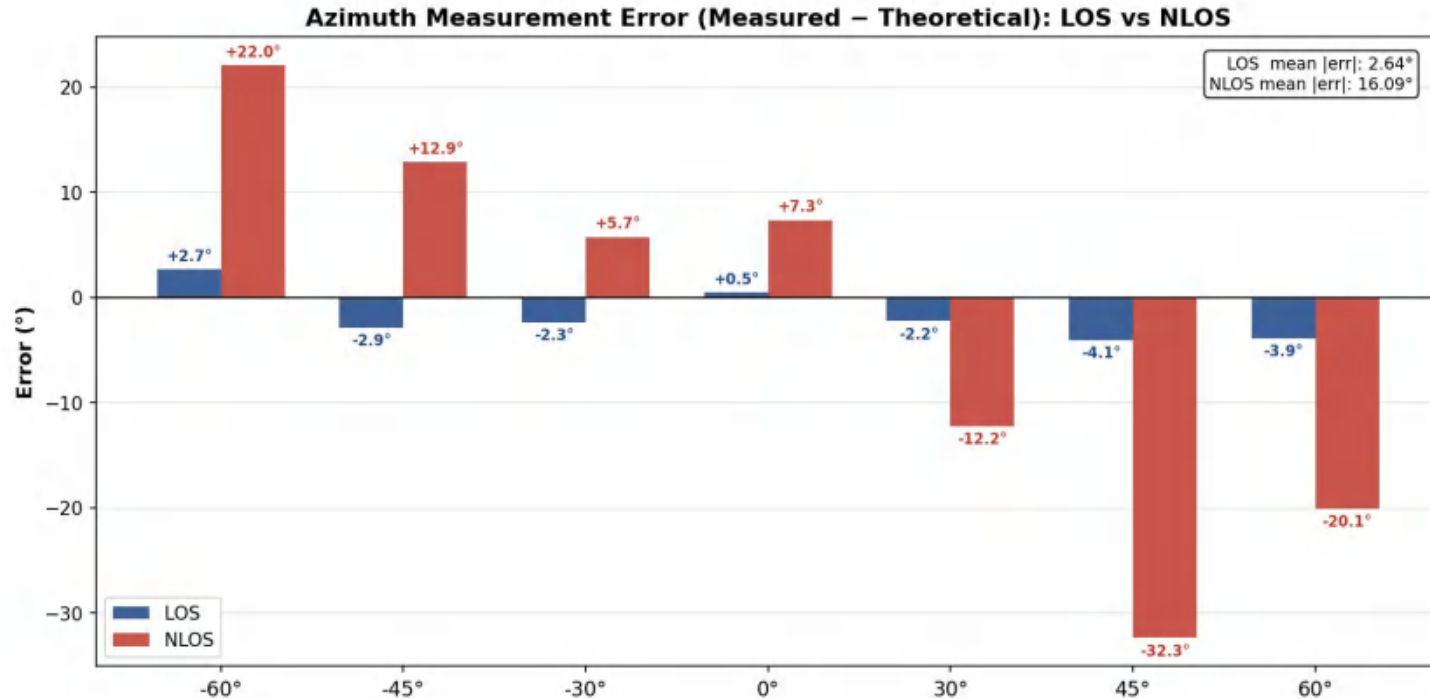
- The LOS standard deviation is generally below 3.5°, indicating stable measurements
- The standard deviation under NLOS is 2–5 times larger than under LOS, significantly increasing measurement instability.

BLE azimuth LOS/NLOS test



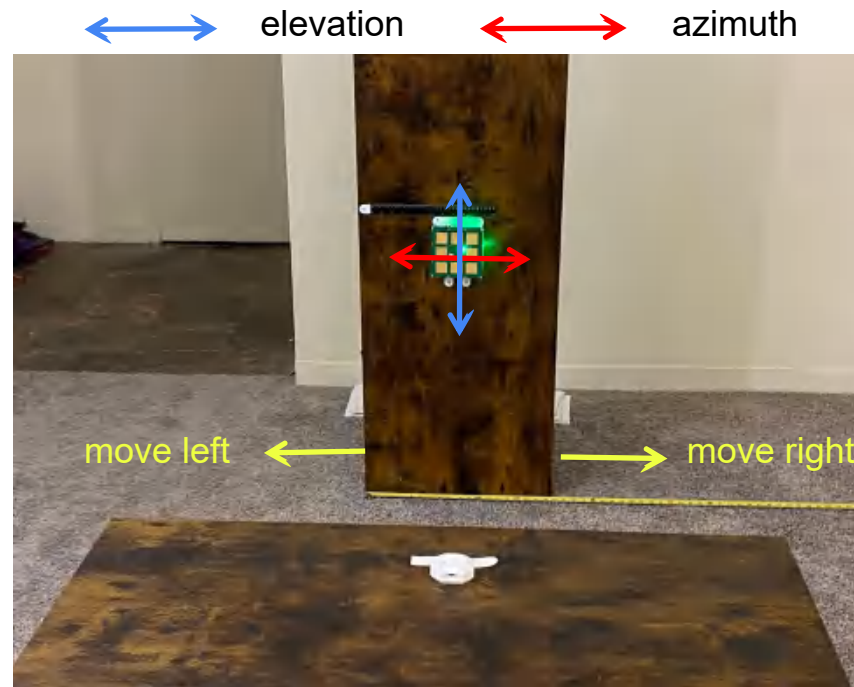
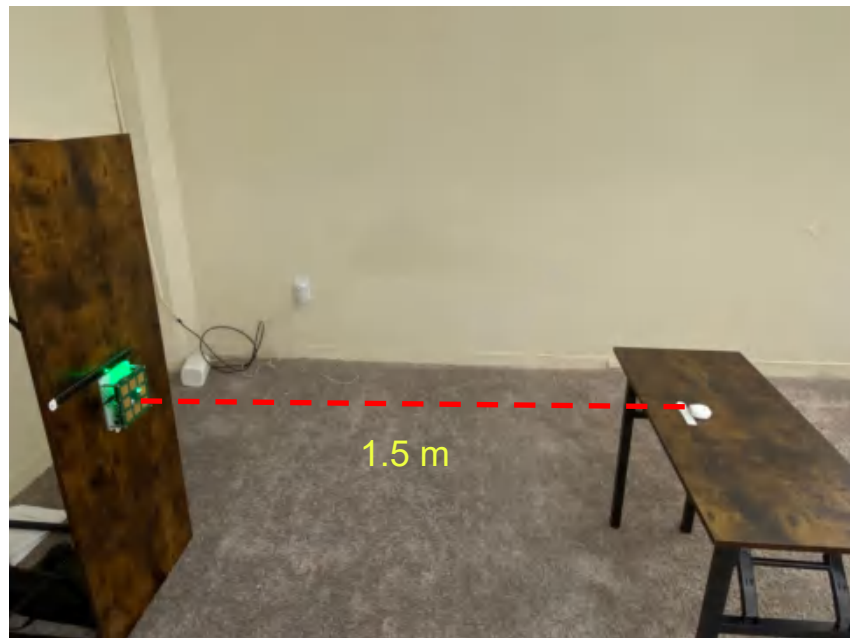
- Under NLOS conditions, the measured values **shift toward 0°** for both positive and negative angles

BLE azimuth LOS/NLOS test



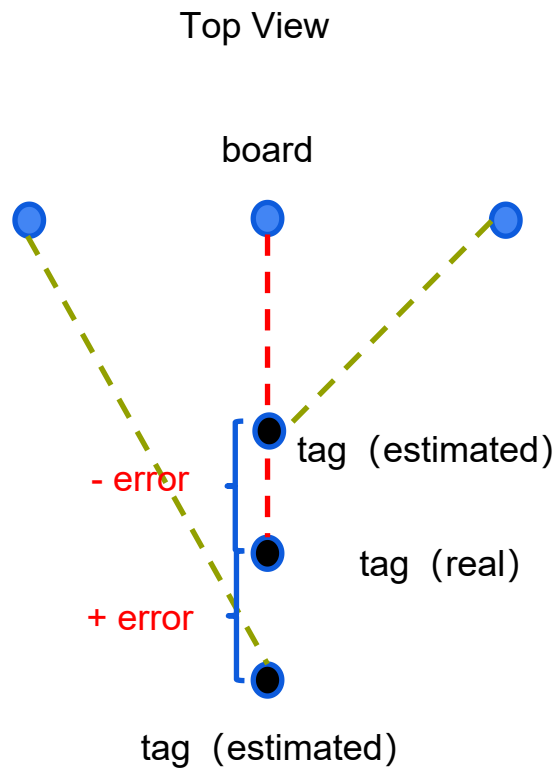
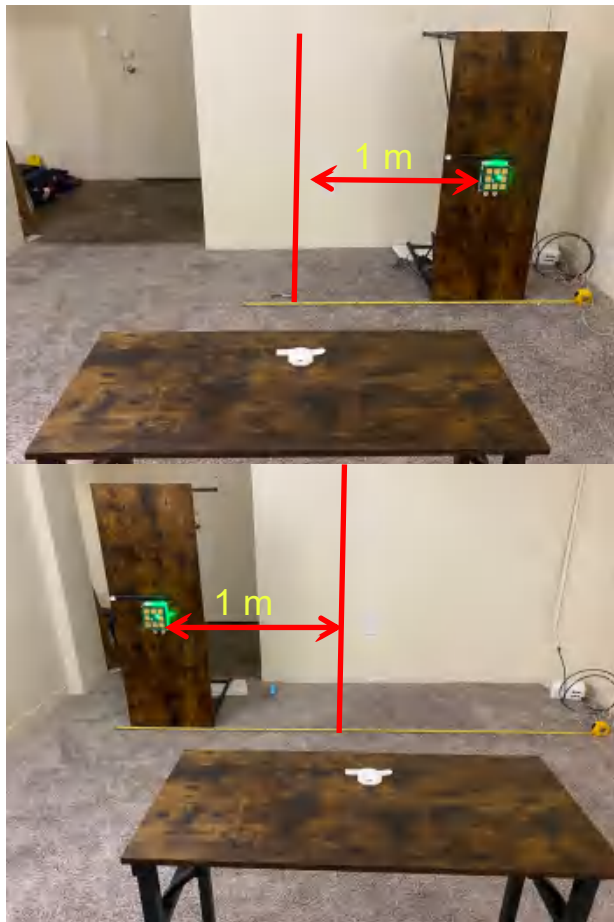
- LOS accuracy is very high, with an average error of only **2.64°**, and the measured values closely match the theoretical values
- The NLOS error is significantly larger; however, when measuring azimuth, the NLOS error does not increase with the angle (It may be related to the orientation of the obstruction)

Setup for LocalizationTest

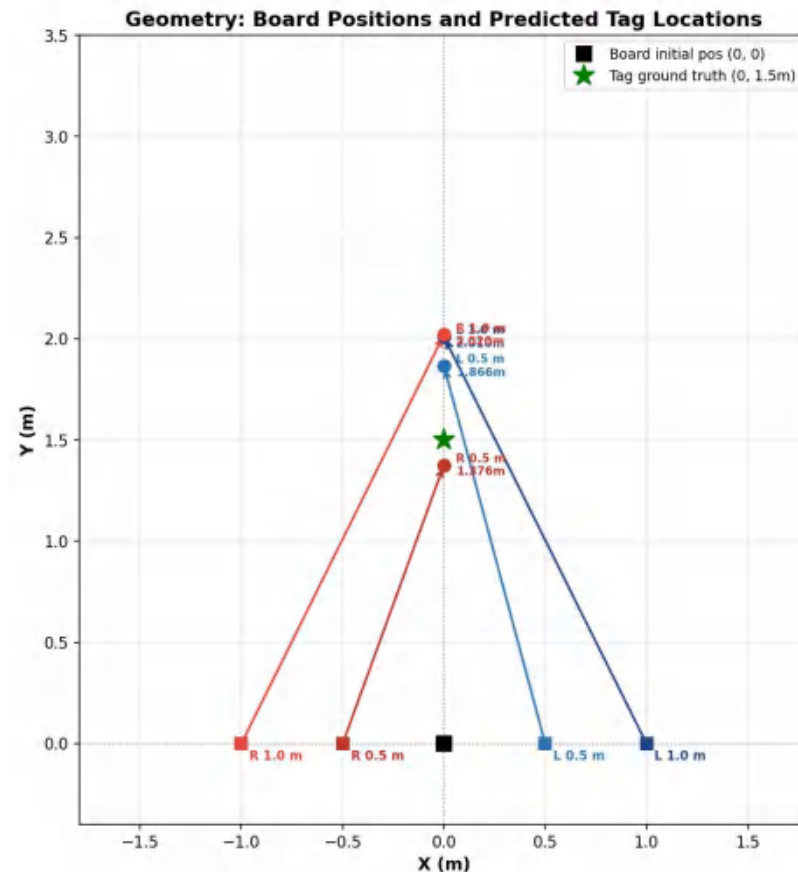
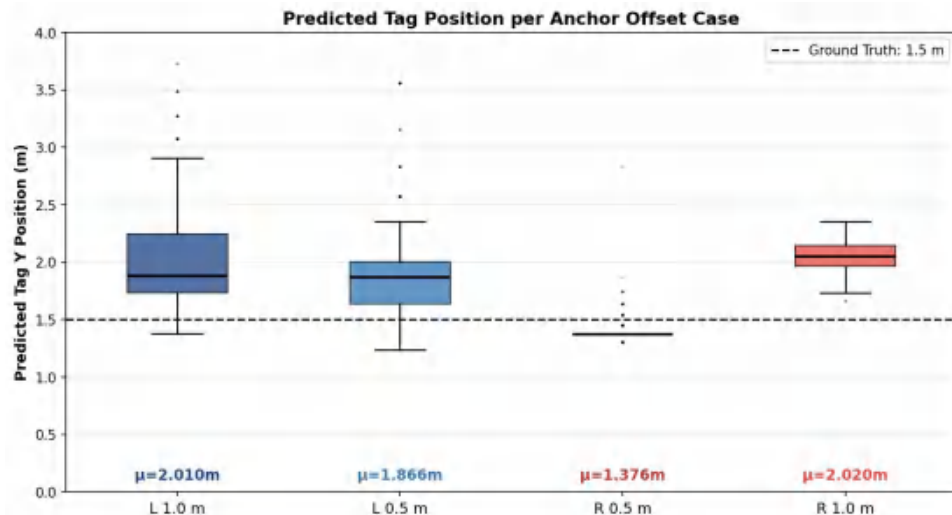


- Fix the **elevation angle** at around 0°
- Define the tag's ground truth position as **1.5 m** directly in front of the antenna array's **initial position**

Setup for Localization Test

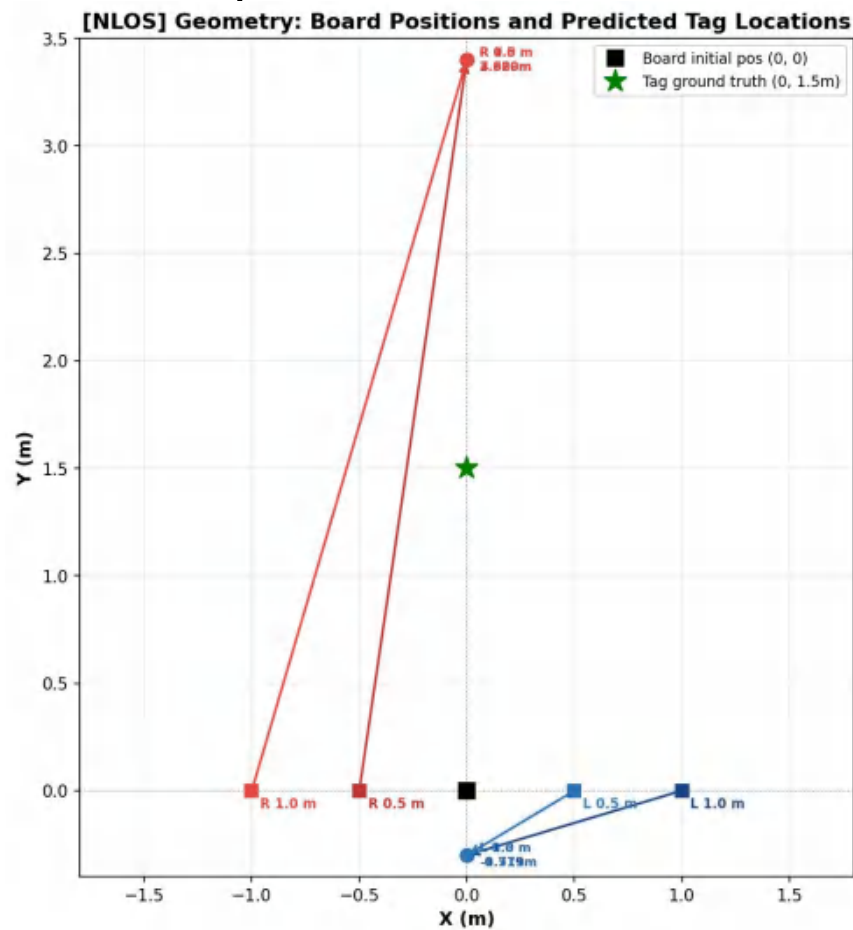
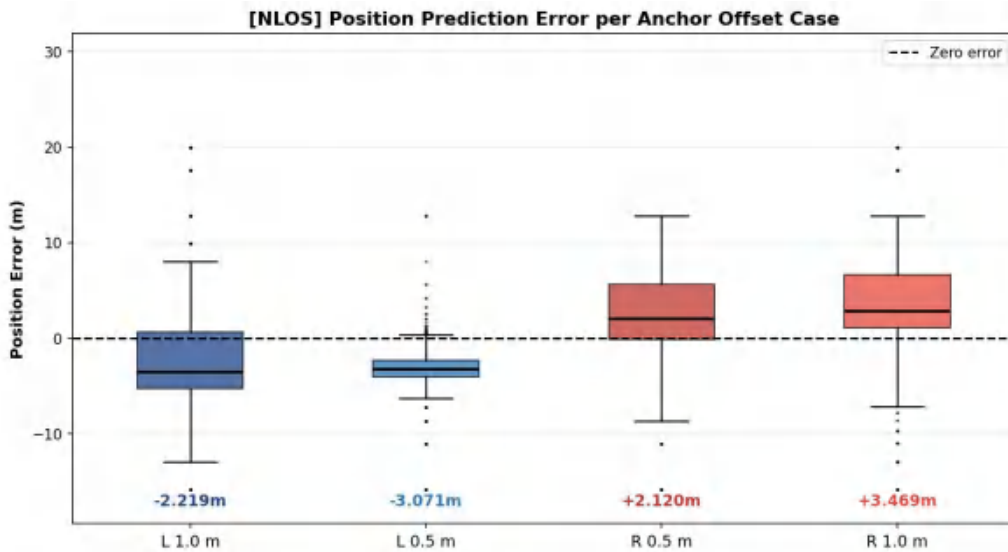


Localization LOS Test Result (Azimuth)



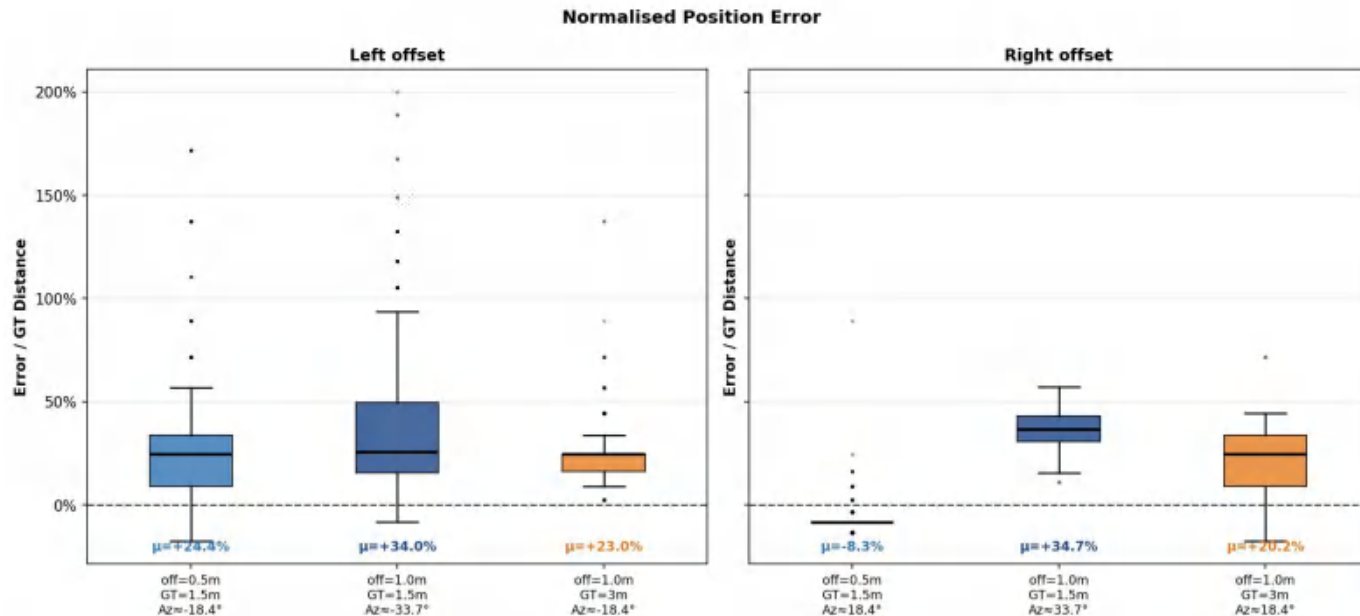
- Right move 0.5m performs best, achieving an MAE of 0.133 m, with a prediction of 1.376 m closest to the 1.5 m ground truth
- Errors for both L and R at 1.0 m are above 0.5 m

Localization NLOS Test Result (Azimuth)



- In NLOS conditions, the azimuth is heavily **biased toward 0°** , leading to large errors in the estimation relative to the ground truth

Localization NLOS Test Result (Azimuth)



- To determine whether the larger errors at ± 1 m are due to the **board's proximity to the wall** or the **increased measurement angles**, tests were conducted with a 3 m ground truth and ± 1 m lateral displacements
- With identical geometric ratios (off/GT), the normalized errors are comparable (L: **25.5% vs 23.0%**, R: **8.9% vs 21.1%**), suggesting that localization error is mainly **angle-dependent** and only weakly related to absolute distance
- The larger variation on the R side is likely due to the R = 0.5 m case having an unusually low standard deviation (0.74°), reflecting superior measurement quality

ROS2 Package: Build & Deploy

Package Structure

```
qingyuan@qingyuan-C7-7588:~$ find -/ros2_iot/src -type f | sort
/home/qingyuan/ros2_iot/src/xplr_aoa3/launch/aoa_launch.py
/home/qingyuan/ros2_iot/src/xplr_aoa3_msgs/CMakeLists.txt
/home/qingyuan/ros2_iot/src/xplr_aoa3_msgs/msg/AoaMeasurement.msg
/home/qingyuan/ros2_iot/src/xplr_aoa3_msgs/package.xml
/home/qingyuan/ros2_iot/src/xplr_aoa3/package.xml
/home/qingyuan/ros2_iot/src/xplr_aoa3/resources/xplr_aoa3
/home/qingyuan/ros2_iot/src/xplr_aoa3/setup.cfg
/home/qingyuan/ros2_iot/src/xplr_aoa3/setup.py
/home/qingyuan/ros2_iot/src/xplr_aoa3/xplr_aoa3/aoa_node.py
/home/qingyuan/ros2_iot/src/xplr_aoa3/xplr_aoa3/__init__.py
qingyuan@qingyuan-C7-7588:~$
```

The complete file structure of two manually created ROS2 packages, including the custom **message definition** (AoaMeasurement.msg), **node code** (aoa_node.py), and **launch files**

Build & Launch Output

```
qingyuan@qingyuan-C7-7588:~$ cd -/ros2_iot
colcon build --packages-select xplr_aoa3
source install/setup.bash
sudo chmod a+r /dev/ttyUSB2
ros2 launch xplr_aoa3 aoa_launch.py port=/dev/ttyUSB2
starting >>> xplr_aoa3
finished <<< xplr_aoa3 [0.88s]

Summary: 1 package finished [1.15s]
[INFO] [launch]: All log files can be found below /home/qingyuan/.ros/log/2
026-04-12-15-26-04-037472-qingyuan-C7-7588-242533
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [aoa_node-1]: process started with pid [242534]
[aoa_node-1] [INFO] [1776021964.396367141] [xplr_aoa3_node]: xplr_aoa3_node
started | port=/dev/ttyUSB2 baud=115200 frame_id=aoa_anchor
[aoa_node-1] [INFO] [1776021964.396749749] [xplr_aoa3_node]: Connected to /
dev/ttyUSB2
[aoa_node-1] [INFO] [1776021974.381829074] [xplr_aoa3_node]: Published 205
measurements total
[aoa_node-1] [INFO] [1776021984.382574478] [xplr_aoa3_node]: Published 397
measurements total
^C[WARNING] [launch]: user interrupted with ctrl-c (SIGINT)
[INFO] [aoa_node-1]: process has finished cleanly [pid 242534]
```

The ROS2 node was successfully **compiled and launched**, connected to the antenna board via **/dev/ttyUSB2**, publishing at approximately 20 Hz, which matches the default broadcast interval (50 ms) of the XPLR-AOA-3

Live Data Verification in ROS

ROS2 Topic Echo — /aoa/measurement

```
*C@ngyuan@ngyuan-G7-7388:~/ros2$ ros2 topic echo /aoa/measurement
header:
  stamp:
    sec: 1776021979
    nanosec: 595431589
  frame_id: aoa_anchor
  tag_id: 20BA360AC7A1
  anchor_id: 20BA369AFC75
  azimuth: 9
  elevation: 18
  rssi: -59
  channel: 35
  timestamp: 1342778
  seq: 48802
  ...
header:
  stamp:
    sec: 1776021979
    nanosec: 859457147
  frame_id: aoa_anchor
  tag_id: 20BA360AC7A1
  anchor_id: 20BA369AFC75
  azimuth: 10
  elevation: 18
  rssi: -68
  channel: 12
  timestamp: 1342828
  seq: 48803
  ...
header:
  stamp:
    sec: 1776021979
    nanosec: 708181632
  frame_id: aoa_anchor
  tag_id: 20BA360AC7A1
  anchor_id: 20BA369AFC75
  azimuth: 10
  elevation: 18
  rssi: -62
  channel: 17
  timestamp: 1342878
  seq: 48804
  ...
header:
  stamp:
    sec: 1776021979
    nanosec: 804822972
  frame_id: aoa_anchor
  tag_id: 20BA360AC7A1
  anchor_id: 20BA369AFC75
  azimuth: 18
```

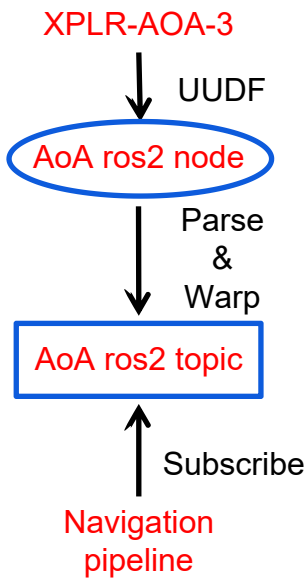
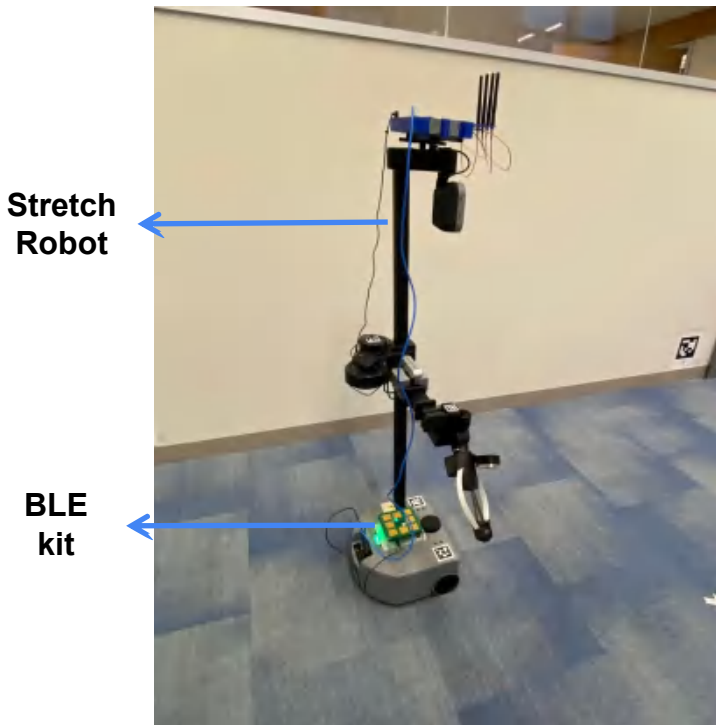
Linux Serial Data Collection

```
[15:00:58.798] Tag: 20BA360AC7A1 | RSSI: -58 dBm | Az: 7° | El: 18°
[15:00:58.791] Tag: 20BA360AC7A1 | RSSI: -61 dBm | Az: 7° | El: 18°
[15:00:58.485] Tag: 20BA360AC7A1 | RSSI: -62 dBm | Az: 7° | El: 18°
[15:00:58.887] Tag: 20BA360AC7A1 | RSSI: -57 dBm | Az: 7° | El: 13°
[15:00:58.898] Tag: 20BA360AC7A1 | RSSI: -52 dBm | Az: 7° | El: 11°
[15:00:58.822] Tag: 20BA360AC7A1 | RSSI: -56 dBm | Az: 8° | El: 12°
[15:00:58.874] Tag: 20BA360AC7A1 | RSSI: -58 dBm | Az: 8° | El: 12°
[15:00:58.837] Tag: 20BA360AC7A1 | RSSI: -61 dBm | Az: 8° | El: 12°
[15:00:58.836] Tag: 20BA360AC7A1 | RSSI: -62 dBm | Az: 8° | El: 14°
[15:00:58.846] Tag: 20BA360AC7A1 | RSSI: -66 dBm | Az: 8° | El: 14°
[15:00:58.857] Tag: 20BA360AC7A1 | RSSI: -63 dBm | Az: 8° | El: 14°
[15:00:58.854] Tag: 20BA360AC7A1 | RSSI: -58 dBm | Az: 8° | El: 14°
[15:00:58.885] Tag: 20BA360AC7A1 | RSSI: -62 dBm | Az: 13° | El: 19°
[15:00:58.931] Tag: 20BA360AC7A1 | RSSI: -62 dBm | Az: 11° | El: 19°
[15:00:58.982] Tag: 20BA360AC7A1 | RSSI: -60 dBm | Az: 13° | El: 19°
[15:00:59.036] Tag: 20BA360AC7A1 | RSSI: -62 dBm | Az: 13° | El: 19°
[15:00:59.078] Tag: 20BA360AC7A1 | RSSI: -66 dBm | Az: 14° | El: 15°
[15:00:59.126] Tag: 20BA360AC7A1 | RSSI: -62 dBm | Az: 14° | El: 15°
[15:00:59.174] Tag: 20BA360AC7A1 | RSSI: -61 dBm | Az: 14° | El: 19°
[15:00:59.222] Tag: 20BA360AC7A1 | RSSI: -60 dBm | Az: 14° | El: 19°
[15:00:59.235] Tag: 20BA360AC7A1 | RSSI: -65 dBm | Az: 14° | El: 19°
[15:00:59.234] Tag: 20BA360AC7A1 | RSSI: -76 dBm | Az: 13° | El: 23°
[15:00:59.282] Tag: 20BA360AC7A1 | RSSI: -76 dBm | Az: 15° | El: 20°
[15:00:59.430] Tag: 20BA360AC7A1 | RSSI: -60 dBm | Az: 15° | El: 20°
[15:00:59.479] Tag: 20BA360AC7A1 | RSSI: -61 dBm | Az: 15° | El: 20°
[15:00:59.526] Tag: 20BA360AC7A1 | RSSI: -73 dBm | Az: 15° | El: 20°
[15:00:59.574] Tag: 20BA360AC7A1 | RSSI: -62 dBm | Az: 15° | El: 20°
[15:00:59.672] Tag: 20BA360AC7A1 | RSSI: -76 dBm | Az: 15° | El: 20°
[15:00:59.685] Tag: 20BA360AC7A1 | RSSI: -62 dBm | Az: 15° | El: 19°
[15:00:59.734] Tag: 20BA360AC7A1 | RSSI: -62 dBm | Az: 15° | El: 19°
[15:00:59.731] Tag: 20BA360AC7A1 | RSSI: -61 dBm | Az: 15° | El: 19°
[15:00:59.833] Tag: 20BA360AC7A1 | RSSI: -65 dBm | Az: 15° | El: 19°
[15:00:59.879] Tag: 20BA360AC7A1 | RSSI: -61 dBm | Az: 14° | El: 19°
[15:00:59.927] Tag: 20BA360AC7A1 | RSSI: -76 dBm | Az: 15° | El: 19°
[15:00:59.974] Tag: 20BA360AC7A1 | RSSI: -65 dBm | Az: 14° | El: 19°
[15:00:59.925] Tag: 20BA360AC7A1 | RSSI: -60 dBm | Az: 14° | El: 19°
[15:00:00.013] Tag: 20BA360AC7A1 | RSSI: -68 dBm | Az: 14° | El: 15°
[15:00:00.034] Tag: 20BA360AC7A1 | RSSI: -64 dBm | Az: 14° | El: 19°
[15:00:00.182] Tag: 20BA360AC7A1 | RSSI: -64 dBm | Az: 14° | El: 19°
[15:00:00.231] Tag: 20BA360AC7A1 | RSSI: -63 dBm | Az: 14° | El: 19°
[15:00:00.279] Tag: 20BA360AC7A1 | RSSI: -59 dBm | Az: 14° | El: 19°
[15:00:00.317] Tag: 20BA360AC7A1 | RSSI: -62 dBm | Az: 14° | El: 19°
[15:00:00.375] Tag: 20BA360AC7A1 | RSSI: -66 dBm | Az: 14° | El: 19°
[15:00:00.423] Tag: 20BA360AC7A1 | RSSI: -70 dBm | Az: 14° | El: 19°
[15:00:00.485] Tag: 20BA360AC7A1 | RSSI: -61 dBm | Az: 14° | El: 19°
[15:00:00.533] Tag: 20BA360AC7A1 | RSSI: -62 dBm | Az: 14° | El: 19°
[15:00:00.582] Tag: 20BA360AC7A1 | RSSI: -65 dBm | Az: 14° | El: 19°
[15:00:00.630] Tag: 20BA360AC7A1 | RSSI: -73 dBm | Az: 14° | El: 19°
[15:00:00.678] Tag: 20BA360AC7A1 | RSSI: -76 dBm | Az: 13° | El: 20°
[15:00:00.785] Tag: 20BA360AC7A1 | RSSI: -62 dBm | Az: 13° | El: 20°
[15:00:00.822] Tag: 20BA360AC7A1 | RSSI: -66 dBm | Az: 13° | El: 20°
[15:00:00.885] Tag: 20BA360AC7A1 | RSSI: -65 dBm | Az: 13° | El: 20°
[15:00:00.931] Tag: 20BA360AC7A1 | RSSI: -66 dBm | Az: 13° | El: 19°
```

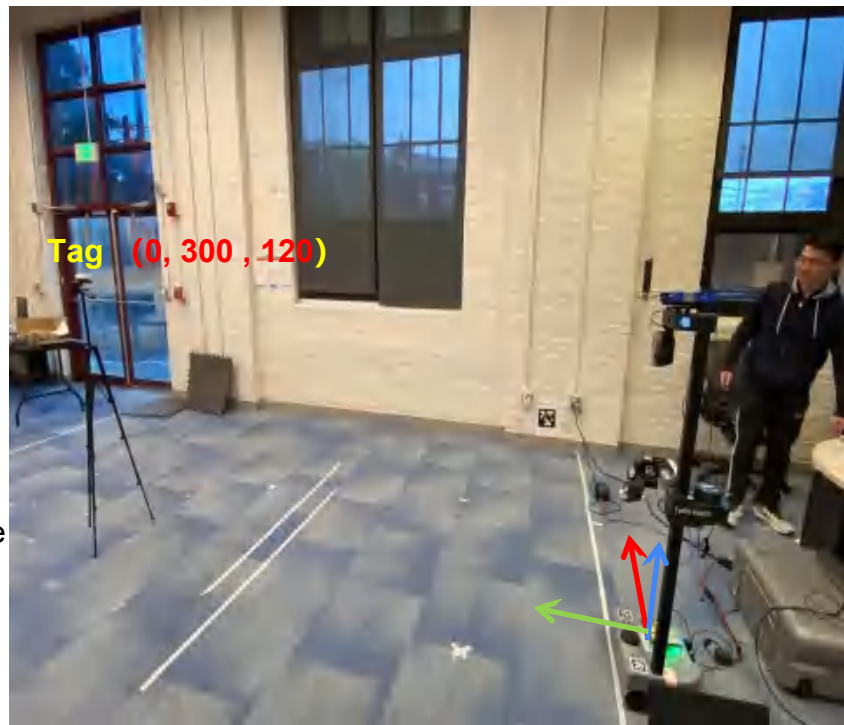
The integrity of the BLE AoA data pipeline was verified at both the serial interface and ROS2 topic levels, demonstrating that the measurement data from the BLE kit can flow end-to-end into the [Linux/ROS2](#) system

Experiment Setup

Integrate the BLE kit with the Stretch Robot



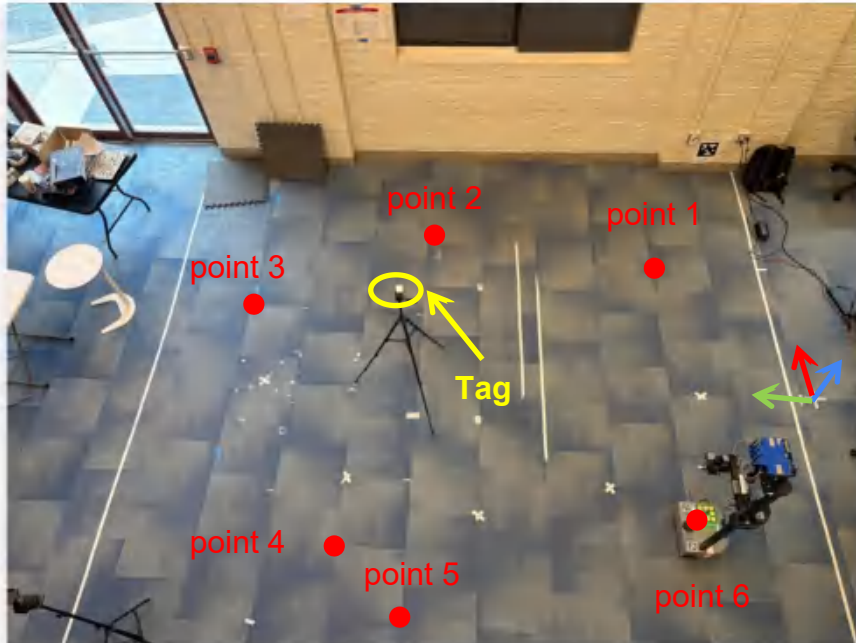
World-frame origin and the tag coordinates



Localization Experiments: LOS vs NLOS

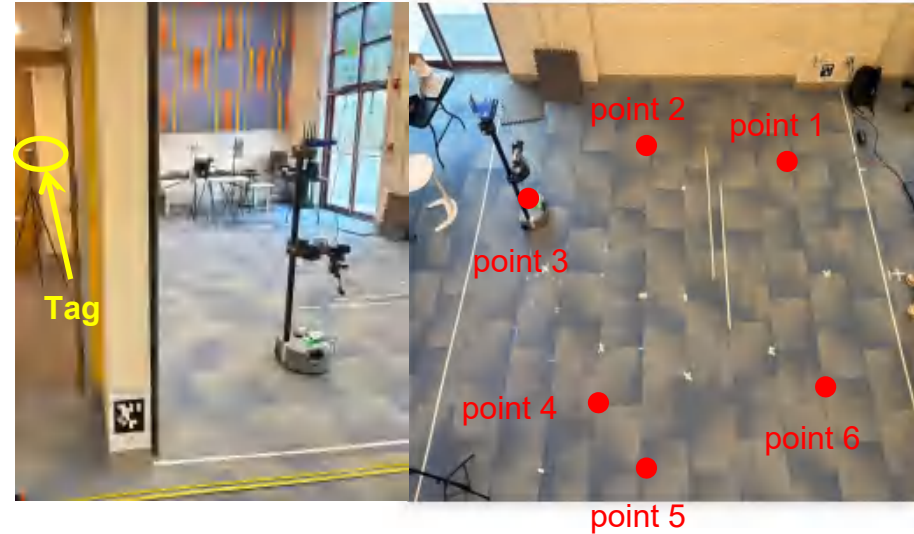
LOS

Top view of the test scenario



NLOS

Separate the robot and the tag with a wall



 [Experiment Video](#)

Tag Localization: LOS vs NLOS

Method

What we already know

position of tag: $\mathbf{p}=(x,y,H)$

position of board: $\mathbf{o}_i=(x_i,y_i,0)$



1. AoA $\rightarrow$ 3D Unit Direction Vector

$$\mathbf{v}_i = \begin{bmatrix} \sin(az_i) \cos(el_i) \\ \cos(az_i) \cos(el_i) \\ \sin(el_i) \end{bmatrix}$$

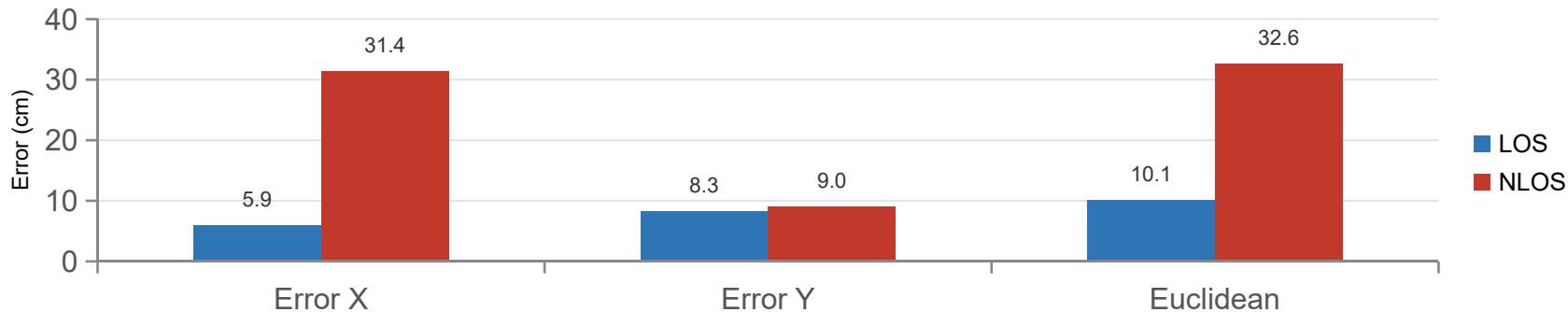


2. Least Squares

$$\min_{\mathbf{p}} \sum_i \|(\mathbf{p} - \mathbf{o}_i) \times \mathbf{v}_i\|^2$$

Results

Condition	Estimated (cm)	Ground Truth (cm)	Error X (cm)	Error Y (cm)	Euclidean (cm)
LOS	(-5.9, 308.3)	(0, 300)	-5.9	+8.3	10.1
NLOS	(-286.6, 294.0)	(-318, 285)	+31.4	+9.0	32.6



AoA Coordinate Conversion

Method

The previous definition of elevation and azimuth **is not suitable** for the current flat board setup

1. AoA → 3D Unit Vector

$$\mathbf{v}_i = \begin{bmatrix} \sin(az_i) \cos(el_i) \\ \cos(az_i) \cos(el_i) \\ \sin(el_i) \end{bmatrix}$$



2. Project → New Az and El

$$az_{new} = \text{atan2}(v_x, v_z)$$
$$el_{new} = \text{atan2}(v_y, \sqrt{v_x^2 + v_z^2})$$

Results

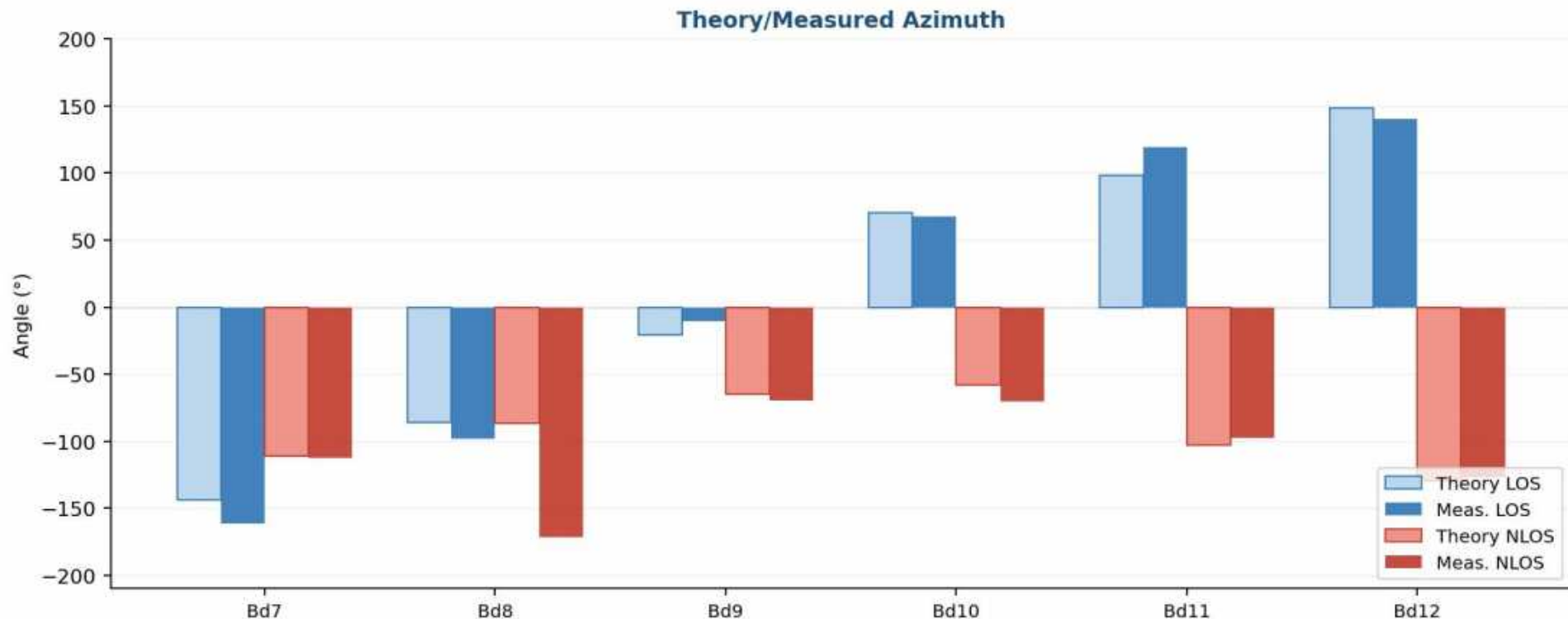
LOS

Point	Az_new (°)	Az Std (°)	El_new (°)	El Std (°)
7	-161.26	3.17	30.14	3.40
8	-98.46	0.83	37.35	1.46
9	-10.58	1.62	35.52	2.85
10	+67.53	1.74	42.03	3.84
11	+118.99	3.62	42.03	6.49
12	+140.58	3.62	49.18	3.72

NLOS

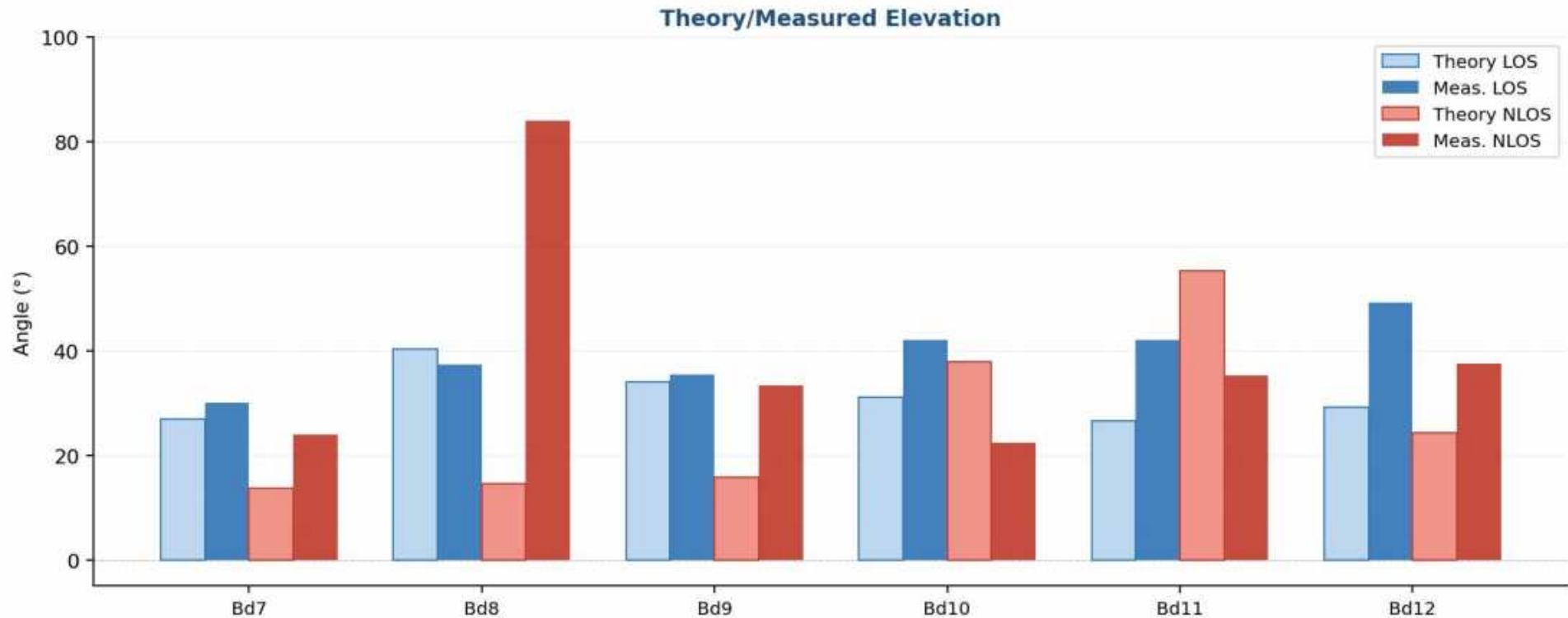
Point	Az_new (°)	Az Std (°)	El_new (°)	El Std (°)
13	-112.72	1.96	23.99	2.2
14	-171.62°	—	84.12°	5.07
15	-69.67	5.01	33.45	5.46
16	-70.47	1.39	22.38	1.85
17	-97.52	1.47	35.29	1.34
18	-126.69	4.16	37.48	7.31

Compare LOS with NLOS

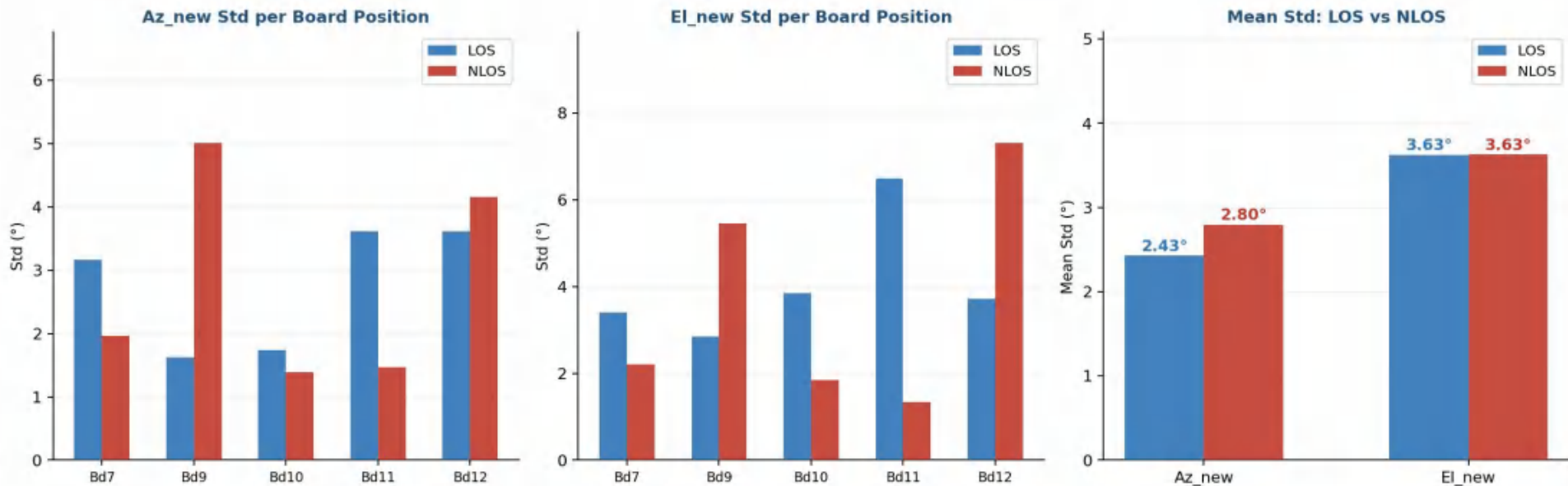


Point 14 is excluded in localization due to measurement failure. The tag's extreme lateral offset (4.59 m) relative to the board height (1.2 m) places it near the $\pm 90^\circ$ previous azimuth boundary, where the antenna array loses angular resolution and produces unreliable readings

Compare LOS with NLOS



Compare LOS with NLOS



- LOS and NLOS show **similar MAE and standard deviation**, but in terms of **localization performance**, **LOS is clearly better than NLOS**
- Under NLOS conditions, the azimuth exhibits a **systematic bias toward 0°**, causing all rays to deviate in the same incorrect direction. So they intersect at a point far from the true tag position
- Under LOS conditions, the **errors are more random**, with rays deviating in different directions. Least squares can partially average out these deviations, leading to a localization result closer to the ground truth

Deliverables

1. BLE Localization pipeline:
 - a. Setup and configuration of the **XPLR-AOA-3 BLE positioning system**
 - b. Ability to collect BLE signal measurements (AoA / RSSI / device identity)
2. Multimodal Signal Data Interface:
 - a. Evaluate the compatibility of XPLR-AOA-3 with Ubuntu
 - b. Package the BLE AoA data stream into a ROS2 node and publish it to ROS2 topics for use by the navigation system
3. Experiment evaluation (If time allows):
 - a. Experiments comparing Wifi and BLE sensing
 - b. Metrics including localization accuracy/AoA error
4. Documentation and Demonstration:
 - a. Setup guide for the XPLR-AOA-3 system for localization
 - b. Example dataset collected from BLE sensing
 - c. Demonstration of BLE signal sensing integrated into the robot navigation pipeline (if time allows, if not we will show other types of demonstration)